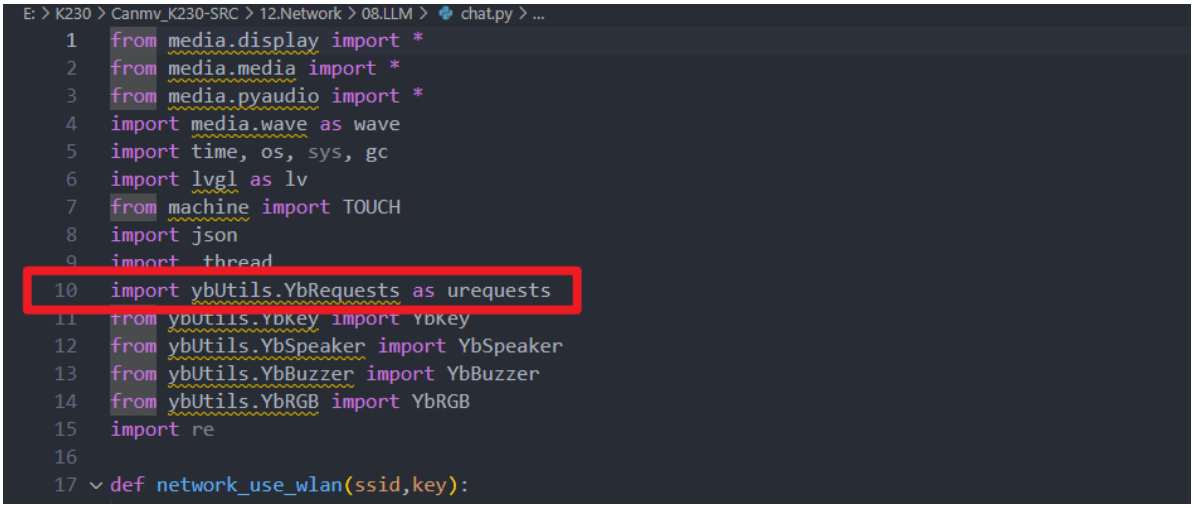
AI语音交互

AI语音交互
- 准备工作
- 例程效果介绍
- 使用方法
- 常见问题

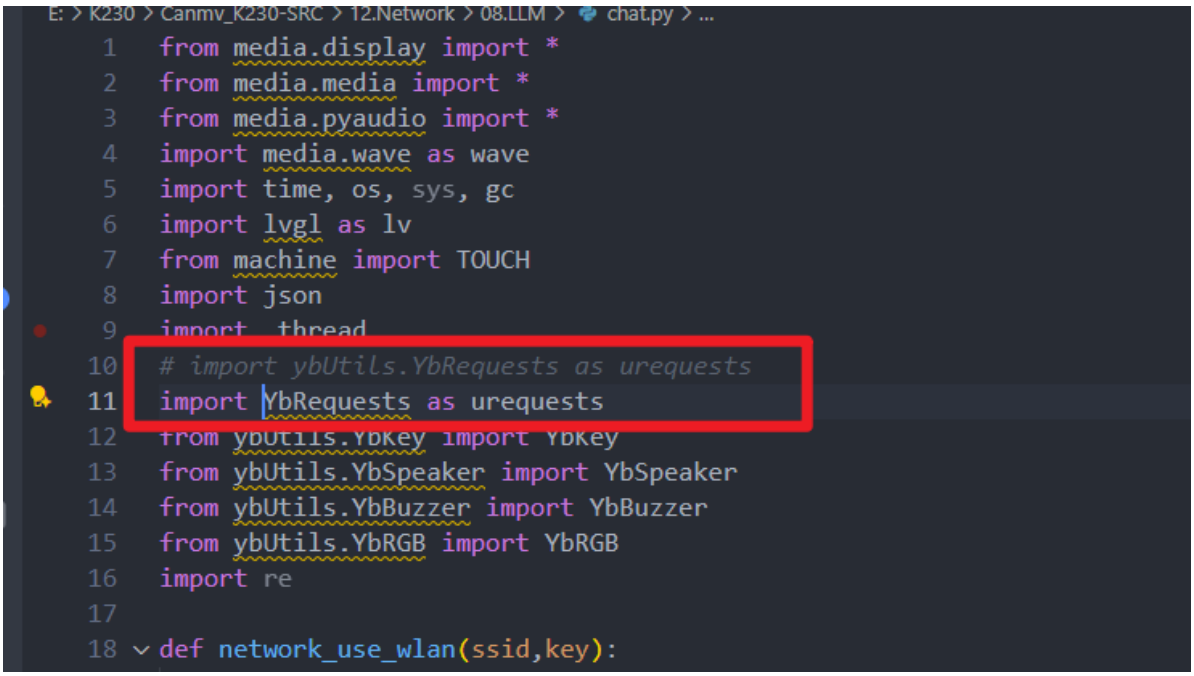
准备工作

若烧录的是1.4.1 及以上版本的固件，需要先从我们官网下载资料包，将 [源码汇总/AI/YbRequests.py] 这个文件放到K230的sdcard目录下，并将例程代码做出下述修改

修改前：



修改后：



例程效果介绍

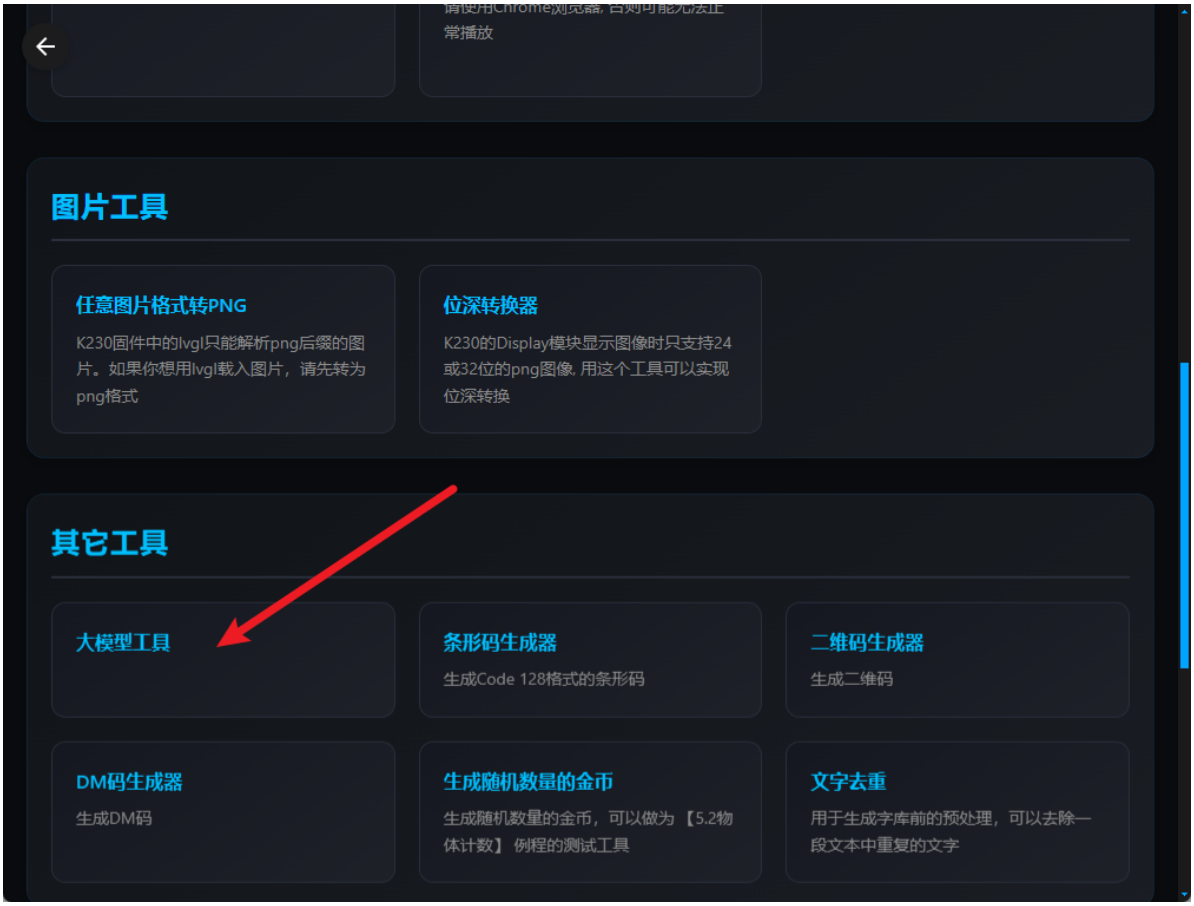
按下K230右上角的Key键后开始录音，松开后结束录音。开始对话

> 注：实际对话效果取决于使用的模型，示例中使用的是Openrouter下的qwen/qwen3-30b-a3b:free



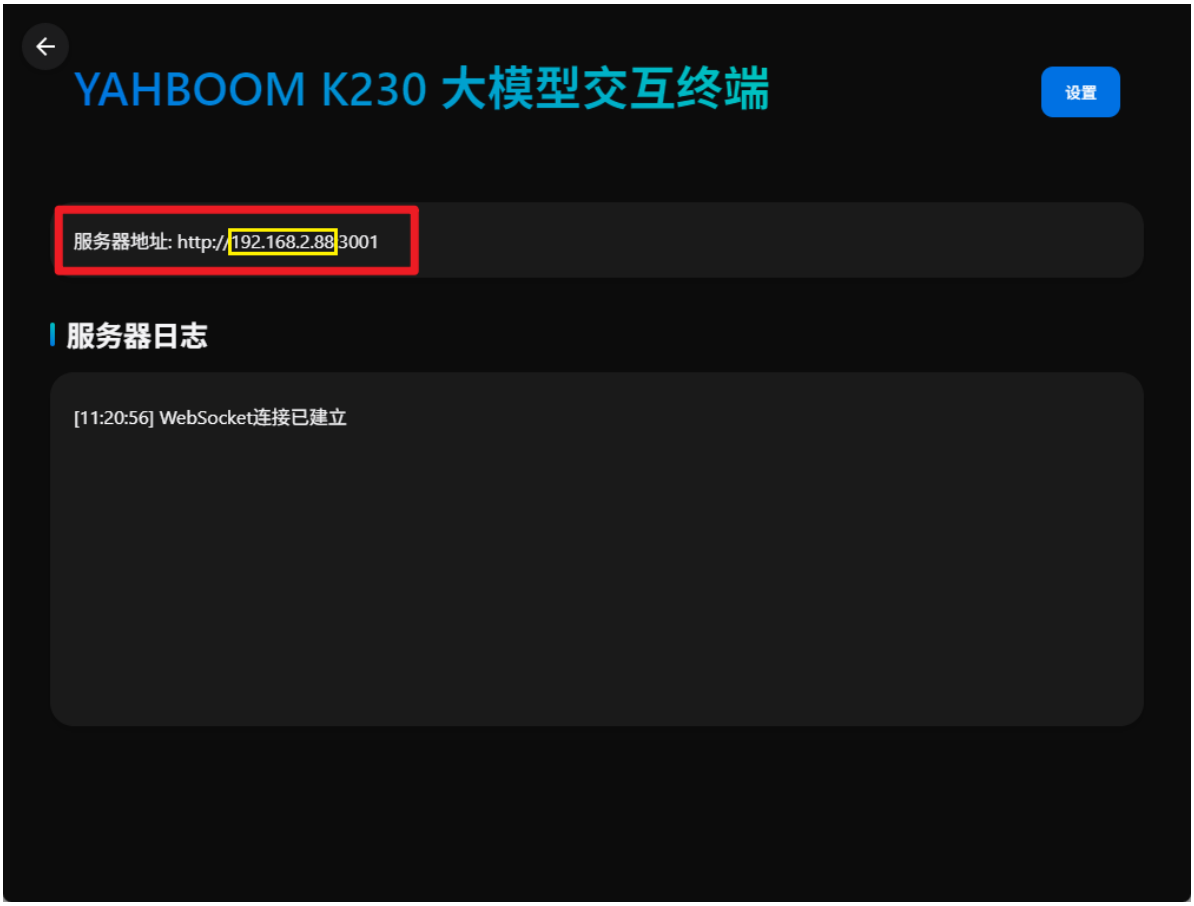
使用方法

打开 K230 配套小工具，找到【大模型工具】，点击启动



复制黄色框里的这一部分地址

> 这里我们需要点击右上角的设置，配置好每一个项目。其中百度的AK和SK用于语音识别和合成，大语言模型相关参数用于实现“思考回答”的过程



打开【源码汇总 / 11.Network / 08.LLM / chat.py】

1. 修改wifi信息 [改为你的wifi名称和密码]

```
# 在1020行附近
def main():
    pass
    os.exitpoint(os.EXITPOINT_ENABLE)
    try:
        ############### 修改这里 ##############################
        print(network_use_wlan("your_wifi_ssid", "your_password"))
        ######################################################
        display_init()
        lvgl_init()
        app = create_chat_app(recorder=None)
        _thread.start_new_thread(voice_serv, ())
```

2. 修改IP地址

```
# 在1046行附近
# 初始化音频流 / Initialize audio stream
rgb = YbRGB()
buzzer = YbBuzzer()
# uart = None
DISPLAY_WIDTH = ALIGN_UP(640, 16)
DISPLAY_HEIGHT = 480

######## 修改这里 ##########
IP = "192.168.2.88"
#########################

chat_url = f"http://{IP}:3001/chat"
chat_app = None
recorder = AudioRecorder()
voice_text_ret = None
```

修改好后点击左下角运行按钮，正常启动输出如下

```
....
Voice service started.
(16895, 6103000, 0, 0, 4939480, 0, 0, 4939480, 0, 6103000)
目录已存在: /data
已创建目录: /data/audio
正在初始化录音组件 ...
初始化完成，等待按键开始录制...
```

> 如果出现run connect failed，请看【常见问题】部分

屏幕显示如图



这时候，我们按下右上角的 Key 键，K230就会开始录音了

松开Key键后开始识别

输出如下（STEP1）

```
录音已保存到: /sdcard/rec.wav
开始语音识别请求: http://192.168.2.88:3001/voice-to-text

[DEBUG] Starting request: POST http://192.168.2.88:3001/voice-to-text
[DEBUG] Parsed URL: {'path': '/voice-to-text', 'query': '', 'scheme': 'http',
'hostname': '192.168.2.88', 'port': 3001}
[DEBUG] Final headers: {'Host': '192.168.2.88', 'Connection': 'close', 'User-
Agent': 'MicroPython/1.0'}
[DEBUG] File upload path: /voice-to-text
[DEBUG] Resolved address: ('192.168.2.88', 3001)
```

> 如果卡在这里超过60秒，请尝试重启K230以及小工具，并确认电脑和K230处于同一网络环境

STEP2

```
[DEBUG] Sent 1024 bytes of file data
[DEBUG] Sent 2048 bytes of file data
[DEBUG] Sent 3072 bytes of file data
[DEBUG] Sent 4096 bytes of file data
[DEBUG] Sent 5120 bytes of file data
[DEBUG] Sent 6144 bytes of file data
[DEBUG] Sent 7168 bytes of file data
[DEBUG] Sent 8192 bytes of file data
[DEBUG] Sent 9216 bytes of file data
```

```
[DEBUG] Sent 10240 bytes of file data
[DEBUG] Sent 11264 bytes of file data
[DEBUG] Sent 12288 bytes of file data
[DEBUG] Sent 13312 bytes of file data
[DEBUG] Sent 14336 bytes of file data
[DEBUG] Sent 15360 bytes of file data
[DEBUG] Sent 16384 bytes of file data
[DEBUG] Sent 17408 bytes of file data
[DEBUG] Sent 18432 bytes of file data
[DEBUG] Sent 19456 bytes of file data
[DEBUG] Sent 20480 bytes of file data
[DEBUG] Sent 21504 bytes of file data
[DEBUG] Sent 22528 bytes of file data
[DEBUG] Sent 23552 bytes of file data
[DEBUG] Sent 24576 bytes of file data
[DEBUG] Sent 25600 bytes of file data
[DEBUG] Sent 26624 bytes of file data
[DEBUG] Sent 27648 bytes of file data
[DEBUG] Sent 28672 bytes of file data
[DEBUG] Sent 29696 bytes of file data
[DEBUG] Sent 30720 bytes of file data
[DEBUG] Sent 31744 bytes of file data
[DEBUG] Sent 32768 bytes of file data
[DEBUG] Sent 33792 bytes of file data
[DEBUG] Sent 34816 bytes of file data
......
具体长度受录制音频长度的影响
......
[DEBUG] Sent 97280 bytes of file data
[DEBUG] Sent 97324 bytes of file data
[DEBUG] Received 405 bytes
[DEBUG] Total response size: 405 bytes
[DEBUG] Response status code: 200
[DEBUG] Closing file upload connection
语音识别成功: 今天天气怎么样？
```

STEP3

输出如下

```
开始AI对话请求: http://192.168.2.88:3001/chat

[DEBUG] Starting request: POST http://192.168.2.88:3001/chat
[DEBUG] Parsed URL: {'path': '/chat', 'query': '', 'scheme': 'http', 'hostname':
'192.168.2.88', 'port': 3001}
[DEBUG] Custom headers: {'Content-Type': 'application/json'}
[DEBUG] Final headers: {'User-Agent': 'MicroPython/1.0', 'Host': '192.168.2.88',
'Content-Type': 'application/json', 'Connection': 'close'}
[DEBUG] Body length: 71
[DEBUG] Full path: /chat
[DEBUG] Request data length: 143
[DEBUG] Creating socket connection to 192.168.2.88:3001
[DEBUG] Resolved address: ('192.168.2.88', 3001)
[DEBUG] Using HTTP connection
[DEBUG] Received 966 bytes
```

```
[DEBUG] Total response size: 966 bytes
[DEBUG] Response status code: 200
[DEBUG] Closing connection
AI对话成功: 请告诉我您所在的城市，我帮您查询实时天气。
(True,
'\u8bf7\u544a\u8bc9\u6211\u60a8\u6240\u5728\u7684\u57ce\u5e02\uff0c\u6211\u5e2e\
u60a8\u67e5\u8be2\u5b9e\u65f6\u5929\u6c14\u3002')
```

STEP 4

```
开始文字转语音请求: http://192.168.2.88:3001/text-to-speech

[DEBUG] Starting request: POST http://192.168.2.88:3001/text-to-speech
[DEBUG] Parsed URL: {'path': '/text-to-speech', 'query': '', 'scheme': 'http',
'hostname': '192.168.2.88', 'port': 3001}
[DEBUG] Custom headers: {'Content-Type': 'application/json'}
[DEBUG] Final headers: {'User-Agent': 'MicroPython/1.0', 'Host': '192.168.2.88',
'Content-Type': 'application/json', 'Connection': 'close'}
[DEBUG] Body length: 75
[DEBUG] Full path: /text-to-speech
[DEBUG] Request data length: 153
[DEBUG] Creating socket connection to 192.168.2.88:3001
[DEBUG] Resolved address: ('192.168.2.88', 3001)
[DEBUG] Using HTTP connection
[DEBUG] Received 1024 bytes
[DEBUG] Received 1024 bytes
[DEBUG] Received 1024 bytes
[DEBUG] Received 1024 bytes
[DEBUG] Received 1024 bytes
.............
.............
[DEBUG] Received 1024 bytes
[DEBUG] Received 1024 bytes
[DEBUG] Received 1024 bytes
[DEBUG] Received 992 bytes
[DEBUG] Total response size: 149472 bytes
[DEBUG] Response status code: 200
[DEBUG] Closing connection
正在保存音频文件...
正在保存文件到: /sdcard/reply.wav
正在写入文件...
文件已保存到: /sdcard/reply.wav
```

至此，一次完整的对话就结束了。

然后我们等待输出如下内容时，就可以进行下一次的对话了

```
初始化完成，等待按键开始录制...
```

常见问题

> 大部分问题可以通过重启PC端的上位机开发工具以及重启K230解决

1. run connect failed
    - 检查WIFI SSID和密码填写是否正确
    - 检查WIFI使用的是否为2.4Ghz
    - 请尝试断电5秒后再重启K230
2. 无法识别语音
    - 检查IP地址是否修改正确
    - 检查PC端和K230是否在同一WIFI环境下
3. 可以识别语音，但是返回：请求错误 -500
    - 使用的大语言模型有问题
        - 如果使用的是讯飞星火，请检查API Key、模型名称是否填写正确，以及使用的API是否有对应模型的权限
    - 如果使用的是Openrouter
        - 请检查API Key、模型名称是否填写正确。如果确认无误，请尝试更换模型使用。Openrouter中的部分模型偶尔会出现这种情况